

Cost aware stopping boundaries in reasoning language models

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When is another reasoning increment worth its cost?

- Repair: an incorrect candidate becomes correct.
- Corruption: a correct candidate becomes incorrect.
- Unproductive continuation: benefit falls below declared cost.
- A causal stop uses the observed prefix and a declared answer selector.

Two corpora answer different questions

Quantity	Revision matrix	Detector corpus
Model-domain cells	52	52
Model configurations	13	13 (different selection)
Trajectories	75,965	28,888
Step rows	798,770 raw	144,440
Unique tasks	1,948	2,948
Horizon	8 / 10 / 14 by domain	5 in every cell

Overlapping corpora; counts are not additive.

The observation unit is a complete revision response

Setting	Declared value
Historical seed / temperatures	7 / 0.1, 0.6, 1.0
Live generator	Local Qwen2.5-0.5B, greedy
Live completion cap	128 tokens per response
Live floor / horizon	2 / 5 responses

The floor and horizon are declared protocol constraints.

Correctness is hidden from the stopping policy

Runtime permits

Current / prior answer and thought

Strict parsing and observed confidence

Charged tokens and public domain

Evaluation keeps separate

Reference answer and correctness

Future response and future label

Bidirectional full-trajectory score

The runtime interface excludes gold and future responses; statistical measurability has separate assumptions.

Accuracy, utility and computation are separate outcomes

- $G_t = q_t - K_t$; maximize $E[G\tau]$ over admissible stops.
- A response-cost example uses $K_t = 0.05(t - 1)$.
- Affine stakes give $G_t = (v + p)q_t - p - K_t$.
- Completion tokens, repeated prompts, peers and wall time need separate ledgers.

Conditional repair and corruption yield the exact drift

- $\alpha_t = P(C_{t+1} = 1 \mid C_t = 0, F_t)$.
- $\beta_t = P(C_{t+1} = 0 \mid C_t = 1, F_t)$.
- $\mu_t = (1 - q_t)\alpha_t - q_t\beta_t - c_t$.
- $c_t = E[K_{t+1} - K_t \mid F_t]$; zero denominators have zero mass.

$$E[G_{t+1} - G_t \mid F_t] = (1 - q_t)\alpha_t - q_t\beta_t - c_t$$

Bellman continuation value defines the optimal stop

- $S_N = G_N$; $S_t = \max\{G_t, E[S_{t+1} \mid F_t]\}$.
- $\tau^* = \text{first } t \geq m \text{ with } S_t = G_t$.
- $\Delta_t^* = \mu_t + E[S_{t+1} - G_{t+1} \mid F_t] \geq \mu_t$.
- Nonpositive immediate drift alone does not imply an optimal stop.

$$S_t = \max(G_t, E[S_{t+1} \mid F_t])$$

A first drift crossing is valid under persistent signs

- τ_μ = first admissible time with $\mu_t \leq 0$, otherwise N .
- If all later conditional drifts stay nonpositive pathwise, τ_μ is optimal.
- Proof: telescope the stopped drift sum before and after the crossing.
- A declining population mean is weaker than this assumption.

$$E[G_\sigma - G_\tau] = \sum_s E[(1\{\sigma > s\} - 1\{\tau > s\}) \mu_s]$$

Delayed repair defeats a myopic stopping rule

Time	Correctness q	Cumulative cost	Stop reward
0	0	0.00	0.00
1	0	0.10	-0.10
2	1	0.20	0.80

Immediate drift is negative at zero, yet optimal continuation value is 0.80.

A ranking score is not a calibrated probability

- AUC measures ordering, not agreement with outcome frequencies.
- Class weighting changes the fitted posterior unless corrected.
- A small set of prefix features need not be a sufficient state.
- Independent task calibration supports marginal checks, not certainty at every prefix.

Repeated checking needs a valid sequential target

- A simultaneous upper bound can control false negative-drift stops.
- Its coverage must hold over all decision times for the actual target.
- Conditional probe independence and pre-probe information must be stated.
- Population diagnostics do not certify one task's unseen continuation.

The freeze makes claims traceable

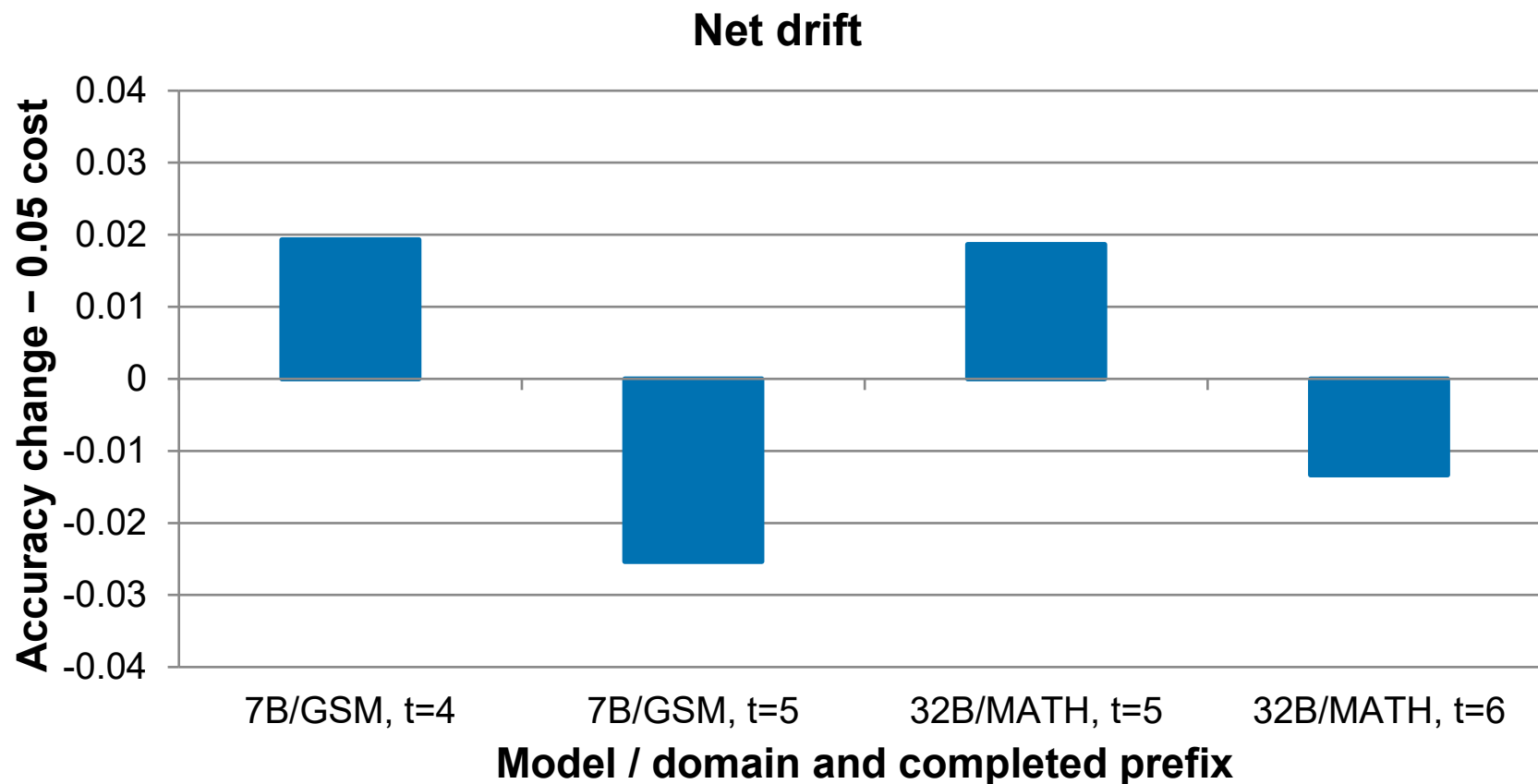
- Exact bytes and canonical LF content have separate hashes.
- Corpus identity, exclusions, partitions and analysis units are explicit.
- Generated proofs are verified by exact finite-system enumeration.
- A hash records identity; it does not certify truth or representativeness.

Repair and corruption use different denominators

Prefix t	Repair / eligible	Corruption / eligible	Net utility drift
2	3145 / 14818	1170 / 4682	+0.0513
4	1830 / 11661	1102 / 7839	-0.0127

This population utility drift can turn negative while average accuracy still increases.

The observed crossing depends on the cell



7B/GSM: crossing 4 → 5.

32B/MATH: crossing 5 → 6.

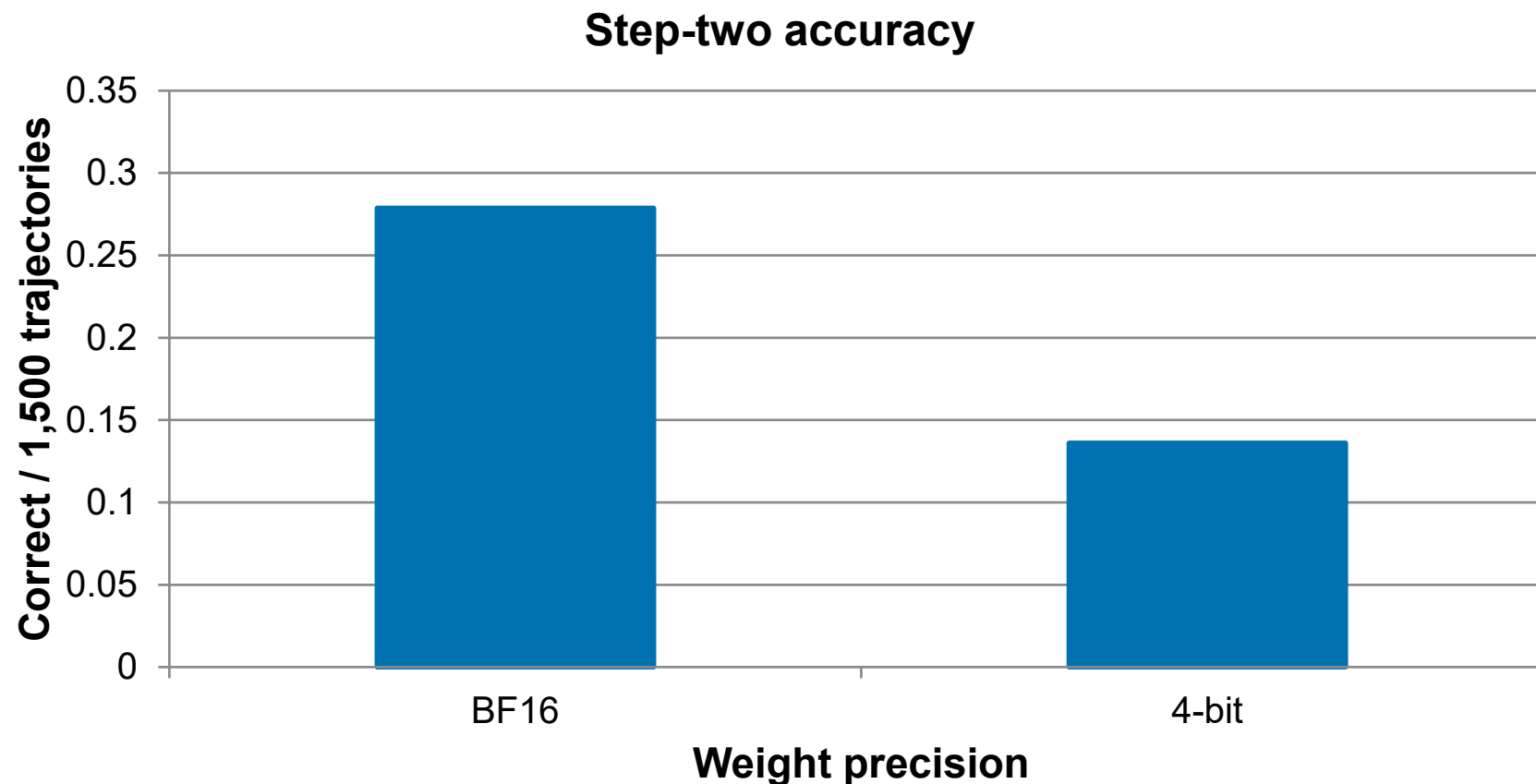
500 tasks per cell;
three temperatures.

More expressive estimators did not always help

Matched modification	Utility / trajectory	95% cell interval
Empirical-Bayes hazards	+0.00781	[+0.00276, +0.01355]
Lagged logistic	+0.00331	[+0.00062, +0.00633]
Gradient-boosted probe	-0.05694	[-0.08716, -0.03145]
Isotonic calibration	-0.06165	[-0.09168, -0.03547]

Matched utility contrasts, with 52-cell descriptive bootstrap intervals.

A paired precision contrast has a large local effect

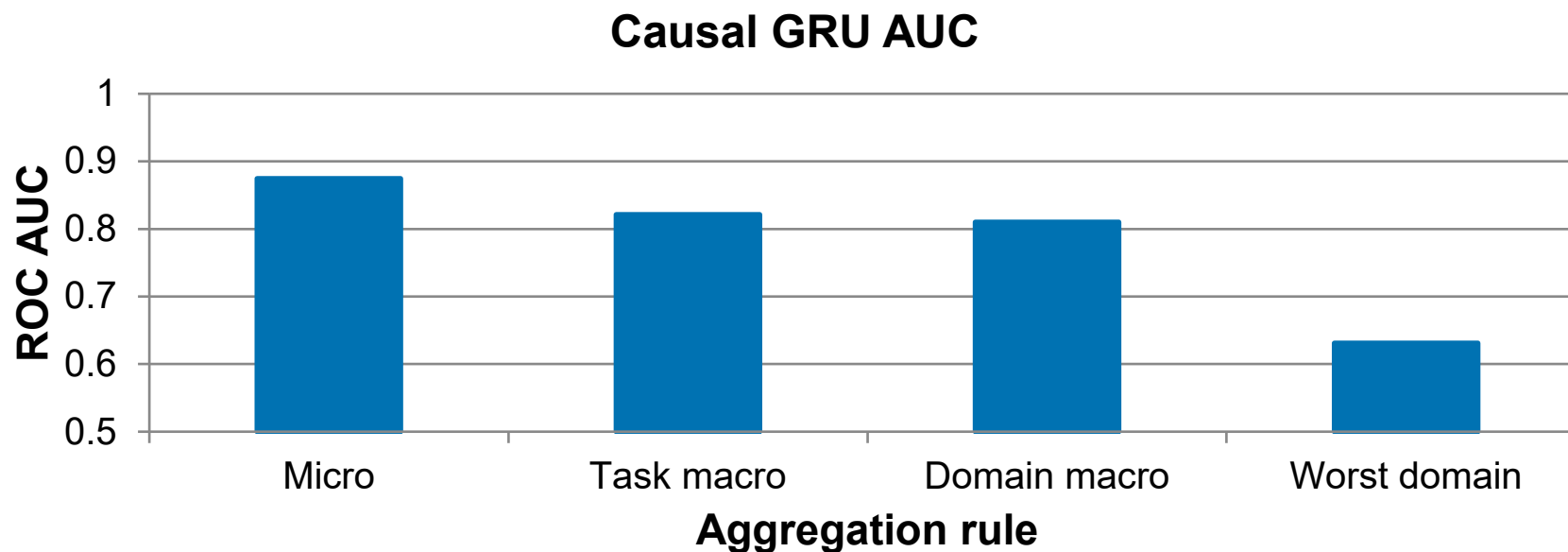


BF16: 418 correct out of 1,500; 4-bit: 204 out of 1,500.

Absolute accuracy difference: 14.27 percentage points.

Task-cluster interval: [11.13, 17.53] points.

Pooled causal discrimination hides a weak domain



GRU micro AUC:
0.8743.

Worst domain GPQA:
0.6313.

AUC and policy utility
rank differently.

Configuration	Micro AUC	Step utility
Causal GRU	0.8743	0.3264
Causal RoPE transformer	0.8638	0.3331

The retrospective 0.955 score is not a live guarantee

Evidence	Permitted claim
Stacked AUC 0.955156	Retrospective development diagnostic
Causal GRU AUC 0.874326	Task-grouped archive prefix ranking
Portable prefix model	Separate train / calibration / holdout
Actual online run	Measured generated-cost / accuracy trade-off
Future-step inputs and non-nested meta-fitting preclude a live 0.955 claim.	

The controller prevents future response calls

Prepared-observation decisions	Latency
Median (saved MATH)	0.0034 ms
95th percentile (saved MATH)	0.0079 ms
99th percentile (saved MATH)	0.0130 ms
Maximum (saved MATH)	0.4383 ms
Learned maximum (4,000 main decisions)	0.9369 ms

Controller timing excludes LM generation/loading, tokenization and peer waits.

A portable trained model estimates current and next correctness

Held-out target	Rows	AUC	Calibrated Brier
Selected current correctness	1380	0.7101	0.0976
Selected next correctness	1104	0.6923	0.0997

Separate task roles; archive calibration does not establish live calibration (0 / 7,500 strict-valid JSON).

The trained drift policy nearly reduces to fixed two

Archive holdout policy	Accuracy	Completion saving	Mean stop
Full horizon	10.51%	0.00%	5.000
Fixed two	11.96%	51.33%	2.000
Learned drift	11.96%	51.29%	2.004

Learned drift nearly equals fixed two; these archived token counts are not measured live savings.

Actual generation yields different policy trade-offs

Actual outcome	Never	Heuristic	Learned
Selected-answer accuracy	6.00%	6.00%	7.00%
Generated completion tokens	22244	21591	9674
Completion-token saving	0%	2.94%	56.51%
Mean stopping step	5.00	4.85	2.00
Shared-prefix identical tasks	Reference	94 / 100	100 / 100

All learned tasks stop at two; added value over fixed two and accuracy noninferiority are not established.

Trap questions probe the policy's failure modes

Trap policy	Accuracy	Completion saving	Paired changes
Heuristic	1 / 20	3.34%	0 worsened / 0 improved
Learned	1 / 20	52.11%	0 worsened / 0 improved

Handpicked bank: descriptive robustness only; all learned tasks stop at two.

The contribution is a testable stopping framework

- Exact conditional drift and finite-horizon optimal-stopping proofs.
- Frozen empirical contrasts retain subgroup and negative results.
- A causal runtime and serialized task-disjoint trained predictor.
- Next: unseen tasks, prespecified accuracy margin, multiple seeds and full costs.